

COMPUTER ORIENTED STATISTICS

Unit-I: Frequency Distribution and Measures of Central Tendency & Dispersion

Comprehensive Study Notes

HPU BCA Semester V

1.1 Introduction to Statistics

What is Statistics?

Statistics is the **science of collecting, organizing, analyzing, interpreting, and presenting data**. It helps in making decisions based on data analysis.

Branches of Statistics

Branch	Description
Descriptive Statistics	Summarizes and describes data (Mean, Median, Mode, Charts)
Inferential Statistics	Draws conclusions from data (Hypothesis testing, Regression)

1.2 Frequency Distribution

Theory

A **Frequency Distribution** is a table that organizes data by showing the number of observations (frequency) in each category or class interval.

1.2.1 Ungrouped Frequency Distribution

Example 1: Marks obtained by 20 students in a test:

25, 30, 35, 25, 40, 35, 25, 30, 35, 40,
30, 25, 35, 40, 30, 25, 35, 30, 40, 35

Marks (x)	Tally Marks	Frequency (f)
25	IIII	4
30	IIII	4
35	IIII I	6
40	IIII	4
Total		20

1.2.2 Grouped Frequency Distribution

Example 2: Ages of 30 employees in a company:

22, 25, 28, 30, 22, 26, 29, 31, 24, 27,
23, 25, 28, 32, 22, 26, 29, 30, 24, 27,
25, 28, 31, 22, 26, 29, 30, 23, 27, 25

Class Interval	Tally Marks	Frequency (f)
20-23	IIII I	6
24-27	IIII IIII	10
28-31	IIII IIII	10

32-35		4
Total		30

Important Terms in Grouped Frequency Distribution

Term	Definition	Example
Class Limits	Boundaries of each class	20-23, 24-27
Class Size (h)	Difference between upper and lower limits	$23 - 20 = 3$
Class Midpoint (x)	Average of class limits	$(20+23)/2 = 21.5$
Cumulative Frequency	Sum of frequencies up to that class	Add as you go down

1.2.3 Construction of Frequency Distribution

Steps:

1. **Find Range:** Maximum value - Minimum value
2. **Decide Number of Classes:** Usually 5-20 classes
3. **Calculate Class Width:** $\text{Range} \div \text{Number of Classes}$ (round up)
4. **Determine Class Limits:** Start with minimum value, add class width
5. **Count Frequencies:** Tally data into classes

1.3 Histogram

Theory

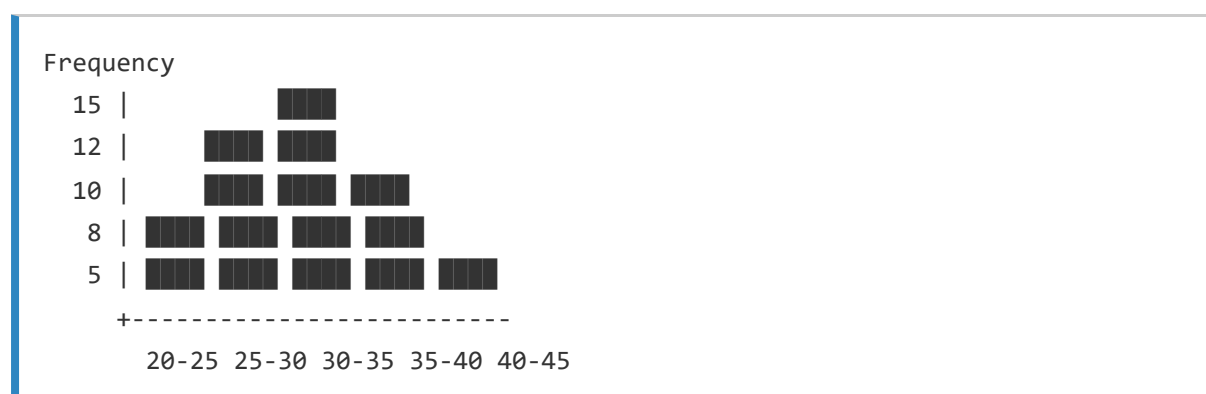
A **Histogram** is a graphical representation of a frequency distribution using **rectangular bars** where:

- **X-axis:** Class intervals
- **Y-axis:** Frequencies
- **Bar width:** Proportional to class width
- **Bar height:** Proportional to frequency

Example: Histogram Construction

Data: Ages of employees

Class	Frequency
20-25	8
25-30	12
30-35	15
35-40	10
40-45	5



Properties of Histogram

1. No gaps between bars (continuous data)

2. Area of bars represents frequency
3. Shows distribution shape (normal, skewed, etc.)
4. Helps in finding mode (highest bar)

Important Formula

For **unequal class intervals**, frequency density is used:

$$\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}$$

1.4 Frequency Polygon

Theory

A **Frequency Polygon** is a line graph created by joining the **midpoints** of the top of each bar in a histogram.

Construction Steps

1. Find **midpoint** of each class: $(LCL + UCL)/2$
2. Plot points: (midpoint, frequency)
3. Connect points with **straight lines**
4. Close the polygon at both ends

Example: Frequency Polygon

Class	Midpoint (x)	Frequency (f)
20-25	22.5	8
25-30	27.5	12
30-35	32.5	15
35-40	37.5	10
40-45	42.5	5

Comparison: Histogram vs Frequency Polygon

Feature	Histogram	Frequency Polygon
Type	Bar chart	Line graph
Points	Rectangles	Points connected
Best for	Distribution shape	Comparing multiple groups
Overlapping	No	Yes (multiple polygons)

1.5 Measures of Central Tendency

Theory

Measures of Central Tendency are values that represent the **center** or **typical value** of a dataset.

1.5.1 Arithmetic Mean

For Ungrouped Data

$$\bar{x} = \frac{\sum x_i}{n}$$

Where: $\bar{x}$ = Mean, x_i = Individual values, n = Number of observations

Example: Find mean of: 5, 8, 10, 12, 15

$$\bar{x} = \frac{5 + 8 + 10 + 12 + 15}{5} = \frac{50}{5} = 10$$

Weighted Mean

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Combined Mean

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

For Grouped Data

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Where: x = Class midpoint, f = Frequency

Example: Calculate mean

Class	f	x (Midpoint)	fx
0-10	5	5	25
10-20	8	15	120
20-30	12	25	300
30-40	6	35	210
40-50	4	45	180
Total	35		835

Mean = $835 / 35 = 23.86$

Step Deviation Method (Shortcut Method)

$$\bar{x} = A + \frac{\sum fd}{n} \times h$$

Where: A = Assumed mean, $d = \frac{x-A}{h}$, h = Class width

Example: Calculate mean using step deviation

Class	f	x	d = (x-A)/h	fd
0-10	5	5	-2	-10
10-20	8	15	-1	-8

20-30	12	25	0	0
30-40	6	35	1	6
40-50	4	45	2	8
Total	35			-4

$$\text{Mean} = 25 + (-4/35) \times 10 = 25 - 1.14 = 23.86$$

1.5.2 Median

For Ungrouped Data

Median is the middle value when data is arranged in ascending/descending order.

Steps:

1. Arrange data in ascending order
2. If n is odd: Median = $\left(\frac{n+1}{2}\right)^{th}$ term
3. If n is even: Median = Average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2} + 1\right)^{th}$ terms

Example 1: Find median of: 3, 7, 9, 12, 15 ($n=5$, odd)

Median = 3rd term = 9

Example 2: Find median of: 3, 7, 9, 12, 15, 18 ($n=6$, even)

Median = $(9 + 12) / 2 = 10.5$

For Grouped Data

$$\text{Median} = L + \frac{\frac{N}{2} - CF}{f} \times h$$

Where: L = Lower boundary of median class, N = Total frequency, CF = Cumulative frequency before median class, f = Frequency of median class, h = Class width

Example: Find median

Class	f	CF
0-10	5	5
10-20	8	13
20-30	12	25
30-40	6	31

40-50	4	35
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$N = 35$, $N/2 = 17.5$
 Median class: 20-30 (where CF crosses 17.5)
 $L = 20$, $CF = 13$, $f = 12$, $h = 10$

 $\text{Median} = 20 + (17.5 - 13)/12 \times 10$
 $= 20 + (4.5/12) \times 10$
 $= 20 + 3.75$
 $= 23.75$

1.5.3 Mode

For Ungrouped Data

Mode is the value that occurs most frequently.

Example: Find mode of: 2, 3, 3, 4, 5, 5, 5, 6

Mode = 5 (appears 3 times)

For Grouped Data

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Where: L = Lower boundary of modal class, f_1 = Frequency of modal class, f_0 = Frequency of class before modal class, f_2 = Frequency of class after modal class, h = Class width

Example: Find mode

Class	f
0-10	5
10-20	8
20-30	12 ← Modal class (highest frequency)

30-40	6
40-50	4

$L = 20, f_1 = 12, f_0 = 8, f_2 = 6, h = 10$

$$\begin{aligned}\text{Mode} &= 20 + (12 - 8) / (2(12) - 8 - 6) \times 10 \\ &= 20 + 4 / (24 - 14) \times 10 \\ &= 20 + (4/10) \times 10 \\ &= 20 + 4 \\ &= 24\end{aligned}$$

Empirical Relationship between Mean, Median, and Mode

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$

1.5.4 Geometric Mean

For Ungrouped Data

$$G = \sqrt[n]{x_1 \times x_2 \times x_3 \times \cdots \times x_n}$$

Using logarithms:

$$\log G = \frac{\sum \log x_i}{n}$$

For Grouped Data

$$G = \left(x_1^{f_1} \times x_2^{f_2} \times \cdots \times x_n^{f_n} \right)^{\frac{1}{N}}$$

$$\log G = \frac{\sum f \log x}{N}$$

Example: Find GM of: 2, 4, 8

$$G = (2 \times 4 \times 8)^{(1/3)} = 64^{(1/3)} = 4$$

Uses of Geometric Mean:

- Growth rates
- Interest rates
- Population growth
- Ratio data

1.5.5 Harmonic Mean

For Ungrouped Data

$$H = \frac{n}{\sum \frac{1}{x_i}}$$

For Grouped Data

$$H = \frac{N}{\sum \frac{f}{x}}$$

Example: Find HM of: 2, 3, 6

$$\begin{aligned} H &= 3 / (1/2 + 1/3 + 1/6) \\ &= 3 / (0.5 + 0.333 + 0.167) \\ &= 3 / 1 \\ &= 3 \end{aligned}$$

Uses of Harmonic Mean:

- Average speed
- Average rate
- Ratios and rates

Comparison of Means

Type	Formula	Best For
Arithmetic Mean	$\bar{x} = \frac{\sum x}{n}$	General average
Geometric Mean	$G = \sqrt[n]{\prod x}$	Growth rates
Harmonic Mean	$H = \frac{n}{\sum \frac{1}{x}}$	Rates and speeds

For same data: $H \leq G \leq \bar{x}$

1.6 Measures of Dispersion

Theory

Measures of Dispersion measure how spread out the data is from the central value.

1.6.1 Range

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Example: 5, 8, 12, 15, 20

$$\text{Range} = 20 - 5 = 15$$

Coefficient of Range:

$$\text{Coef. of Range} = \frac{\text{Max} - \text{Min}}{\text{Max} + \text{Min}}$$

For Grouped Data

$$\text{Range} = \text{Upper limit of last class} - \text{Lower limit of first class}$$

1.6.2 Quartile Deviation

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

Where: Q_1 = First Quartile (25th percentile), Q_3 = Third Quartile (75th percentile)

Coefficient of Quartile Deviation

$$\text{Coef. of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

1.6.3 Mean Deviation

For Ungrouped Data

$$\text{MD} = \frac{\sum |x - \bar{x}|}{n}$$

For Grouped Data

$$\text{MD} = \frac{\sum f|x - \bar{x}|}{\sum f}$$

Example: Find MD of: 2, 4, 6, 8, 10 (Mean = 6)

x	x - 6
2	4
4	2
6	0
8	2
10	4
Total	12

$$\text{MD} = 12 / 5 = 2.4$$

Coefficient of Mean Deviation

$$\text{Coef. of MD} = \frac{\text{Mean Deviation}}{\text{Mean}}$$

1.6.4 Variance

For Ungrouped Data

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Alternative Formula:

$$\sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

For Grouped Data

$$\sigma^2 = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

Example: Find variance of: 2, 4, 6, 8, 10

x	x - $\bar{x}$	$(x - \bar{x})^2$
2	-4	16
4	-2	4
6	0	0
8	2	4

10	4	16
Total		40

Variance = $40 / 5 = 8$

1.6.5 Standard Deviation

$$\sigma = \sqrt{\text{Variance}} = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Example: From previous example

SD = $\text{sqrt}(8) = 2.828$

Properties of Standard Deviation

1. SD is always non-negative
2. SD is independent of change of origin (adding/subtracting constant)
3. SD is affected by change of scale (multiplying/dividing)
4. SD is measured in same units as data

Step Deviation Method for Standard Deviation

$$\sigma = h \times \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}$$

Where $d = \frac{x-A}{h}$

1.6.6 Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

Example: Compare two data sets

Dataset	Mean	SD	CV
A	50	5	10%
B	100	10	10%

Interpretation: Lower CV = Less variable, Higher CV = More variable

1.7 Complete Examples

Example 1: Complete Frequency Distribution Analysis

Data: 50 students' marks in statistics:

45, 52, 38, 61, 48, 55, 42, 58, 50, 43,
56, 49, 62, 47, 53, 44, 57, 40, 51, 46,
59, 41, 54, 60, 39, 63, 55, 48, 52, 44,
58, 50, 46, 61, 47, 53, 42, 56, 49, 45,
54, 40, 57, 51, 43, 62, 55, 48, 50, 46

Step 1: Construct Frequency Distribution

Class	Tally	f	x (mid)	fx	CF
35-40	III	3	37.5	112.5	3
40-45	IIII IIII	8	42.5	340	11
45-50	IIII IIII I	11	47.5	522.5	22
50-55	IIII IIII II	12	52.5	630	34
55-60	IIII IIII	9	57.5	517.5	43
60-65	IIII II	7	62.5	437.5	50
Total		50		2560	

Step 2: Calculate Mean

$$\text{Mean} = 2560 / 50 = 51.2$$

Step 3: Calculate Median

$$N = 50, N/2 = 25$$

Median class: 50-55 (CF crosses 25)

$$L = 50, CF = 22, f = 12, h = 5$$

$$\text{Median} = 50 + (25 - 22)/12 \times 5$$

$$= 50 + 1.25$$

$$= 51.25$$

Step 4: Calculate Mode

Highest frequency: 12 (50-55 class)

$L = 50$, $f_1 = 12$, $f_0 = 11$, $f_2 = 9$, $h = 5$

$$\begin{aligned}\text{Mode} &= 50 + (12 - 11) / (2(12) - 11 - 9) \times 5 \\ &= 50 + 1 / (24 - 20) \times 5 \\ &= 50 + 1.25 \\ &= 51.25\end{aligned}$$

Step 5: Calculate Standard Deviation

Class	f	x	$d = \frac{x-52.5}{5}$	fd	fd ²
35-40	3	37.5	-3	-9	27
40-45	8	42.5	-2	-16	32
45-50	11	47.5	-1	-11	11
50-55	12	52.5	0	0	0
55-60	9	57.5	1	9	9
60-65	7	62.5	2	14	28
Total	50			-13	107

$$\begin{aligned}\text{SD} &= 5 \times \sqrt{107/50 - (-13/50)^2} \\ &= 5 \times \sqrt{2.14 - 0.0676} \\ &= 5 \times \sqrt{2.0724} \\ &= 5 \times 1.4396 \\ &= 7.198\end{aligned}$$

Example 2: Comparison of Dispersion

Data Set A: 10, 20, 30, 40, 50

Data Set B: 90, 100, 110, 120, 130

Measure	Set A	Set B
Mean	30	110
Range	40	40
Variance	200	200
SD	14.14	14.14
CV	47.13%	12.85%

Interpretation: Although SD is same, Set A is more variable (CV higher).

1.8 Key Formulas Summary

Measures of Central Tendency

Measure	Formula	Type
Mean	$\bar{x} = \frac{\sum x}{n}$	Ungrouped
Mean	$\bar{x} = \frac{\sum fx}{\sum f}$	Grouped
Median	Middle value	Ungrouped
Median	$L + \frac{N/2 - CF}{f} \times h$	Grouped
Mode	Most frequent	Ungrouped
Mode	$L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$	Grouped
Geometric Mean	$\sqrt[n]{x_1 x_2 \dots x_n}$	Ungrouped
Harmonic Mean	$\frac{n}{\sum 1/x}$	Ungrouped

Measures of Dispersion

Measure	Formula
Range	Max - Min
Mean Deviation	$\frac{\sum x - \bar{x} }{n}$
Variance	$\frac{\sum (x - \bar{x})^2}{n}$
Standard Deviation	$\sqrt{\frac{\sum (x - \bar{x})^2}{n}}$
Coefficient of Variation	$\frac{\sigma}{\bar{x}} \times 100$

Skewness

Skewness measures the asymmetry of a distribution.

Type	Description	Mean-Median-Mode Relationship
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Positive Skew	Tail to the right	Mean > Median > Mode
Negative Skew	Tail to the left	Mean < Median < Mode
Zero Skew (Symmetric)	Bell-shaped	Mean = Median = Mode

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

Previous Year Questions

Short Questions (2 marks)

- Q1.** Define Frequency Distribution.
- Q2.** What is a Histogram?
- Q3.** Define Mean and Median.
- Q4.** What is the difference between Variance and Standard Deviation?
- Q5.** What is Coefficient of Variation?
- Q6.** Define Geometric Mean.
- Q7.** What is a Frequency Polygon?
- Q8.** What is the empirical relationship between Mean, Median, and Mode?

Medium Questions (5 marks)

- Q1.** What is Frequency Distribution? Explain its types with examples.
- Q2.** Explain the construction of Histogram and Frequency Polygon.
- Q3.** Calculate Mean, Median, and Mode for grouped data.
- Q4.** What are measures of dispersion? Explain any two.
- Q5.** Find the Geometric Mean and Harmonic Mean of 2, 4, 8, 16.

Long Questions (12 marks)

- Q1.** (a) Construct a frequency distribution table for the given data.
(b) Calculate arithmetic mean and median.
- Q2.** (a) What are measures of central tendency? Explain all measures with formulas.
(b) Calculate mean using step deviation method.
- Q3.** (a) What is dispersion? Why is it important?
(b) Calculate standard deviation and coefficient of variation.

Important Formulas to Memorize

1. Mean (Ungrouped): $\bar{x} = \frac{\sum x}{n}$
2. Mean (Grouped): $\bar{x} = \frac{\sum fx}{\sum f}$
3. Median (Grouped): $L + \frac{N/2 - CF}{f} \times h$
4. Mode (Grouped): $L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
5. Variance: $\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$
6. Standard Deviation: $\sigma = \sqrt{\sigma^2}$
7. Geometric Mean: $G = \sqrt[n]{x_1 x_2 \dots x_n}$
8. Harmonic Mean: $H = \frac{n}{\sum 1/x}$
9. CV: $\frac{\sigma}{\bar{x}} \times 100$